Abhinav Kumar

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Professional Experience

PwC, ***Specialist*** – On-site July 2025 – Present

- Built and optimized GenAI-driven NLP pipelines leveraging RAG architectures and OpenAI LLM fine-tuning on custom datasets, reducing API costs by 30% and enhancing response accuracy through domain adaptation and prompt engineering for context-aware, domain-specific intelligence.
- Built **scalable backend services and low-latency decision engines**, optimizing database interactions, service logic, API performance, and downstream integrations.
- Developed reusable **React.js, Angular.js, and TypeScript SSR components** for policy search and submission workflows, improving reusability, performance, and UX.
- Managed **Kubernetes deployments** with horizontal and vertical scaling, and used **Terraform** for consistent, repeatable cloud infrastructure provisioning across environments.
- Automated workflows through **Agentic AI solutions** using LangChain, CrewAI, and FastAPI, reducing repetitive processes.

IBM Phemesoft, ***Software Engineer Intern*** – Remote June 2024 – July 2024

- Architected a serverless data pipeline using **AWS S3, Glue, and ETL jobs** for large-scale data processing of **500+ GB**.
- Queried **100+ GB** of processed data using **AWS Athena** and automated **12+** daily workflows with **AWS Lambda**.
- Implemented **5** interactive **AWS QuickSight dashboards** to provide actionable business insights for **20+** key stakeholders.

Technical Skills

Languages:	Python, Java, C++, SQL
Backend:	REST APIs, Microservices, Spring Boot, Hibernate, Maven, JQuery, JUnit
Cloud & DevOps:	AWS (Lambda, EC2, S3, IAM), Docker, Kubernetes, Terraform, GitHub, CI/CD), Azure
AI & LLM:	LangChain, OpenAI API, Hugging Face, RAG, Crew Ai, FAISS, Ollama, Deep Learning, NLP, LLM
Core CS:	Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

Projects

Cloud File Vault | *Java, React, MySql, AWS S3, RDS, EC2, Lambda, JDBC* Github

- Engineered a secure cloud-based file storage system enabling reliable file management and metadata persistence.
- Implemented OTP-based authentication using the JavaMail API, strengthening user access control and application security.
- Designed scalable storage solutions using AWS S3 and Amazon RDS for secure file management and efficient data retrieval.

Film Seeker | *Python,KNN, Rest API, NLP* Github

- Developed a **movie recommendation system** using **KNN-based similarity** and **NLP** for personalized suggestions.
- Designed and integrated **REST APIs** to fetch and serve movie data, enabling efficient client-server interaction.

StreamForge – Kafka-Compatible Message Broker | *C++, CMake, TCP/IP, Multithreading* Github

- Engineered a **distributed message broker** in modern **C++** supporting persistent partitioned logs, Kafka protocol request/response handling, concurrent client processing, and consumer message fetching.
- Designed a **thread-pool-based request-processing layer** to handle concurrent broker clients while separating networking, protocol parsing, and storage responsibilities.
- Built **persistent log storage and metadata management** with **CMake**-based unit testing, improving broker reliability.

Education

University of Petroleum & Energy Studies - B.Tech in Computer Science & Engineering June 2021 – June 2025

- CGPA: 7.6/10

Certifications & Achievements

- **GeeksforGeeks — Data Structures and Algorithms** View
- **AWS Certified Developer — Associate** View
- **Ranked 67th among 13,000+ participants (Top 0.6%) in a Codeforces competitive programming contest.**
- **Solved 600+ DSA problems** across LeetCode and GeeksforGeeks .